

Vaishnavi V

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Summary

Computer Science Engineering student with a strong foundation in Python, AI, ML, and NLP, along with hands-on experience developing AI-powered, full-stack, and database-driven applications. Skilled in software development, backend services, and integrating technologies to build practical software solutions. Strong analytical and problem-solving abilities with a proven ability to adapt to new technologies and collaborate effectively in team environments. Seeking a Software Engineer role to contribute to scalable, reliable, and maintainable software solutions.

Experience

Academic Research:

- Conducted technical research on emerging technologies to support academic projects.
- Collected, analysed, and summarized information from reliable technical sources.
- Collaborated with teammates to prepare project documentation and presentations.
- Improved problem-solving and analytical skills through research-based learning.

Education

B.E. in Computer Science and Engineering

Shri Madhwa Vadiraja Institute of Technology and Management, Bantakal

CGPA: 7.62 | 2023 - Present

12th Standard (PCMB)

Vidyodaya PU College

88.33% | 2023

10th Standard

Vidyodaya Public School

90.33% | 2021

Skills

- Programming Languages:** Python, Java, C
- AI:** AI, ML, NLP, GenAI
- Libraries:** OpenCV, Tesseract OCR, YOLO
- Web Technologies:** HTML, CSS, JavaScript, React
- Database:** SQL (MySQL), MongoDB
- Tools:** Git, GitHub, Docker, VS Code.

Projects

AI-Based Smart Traffic Congestion Optimization

- Technologies:** Python, OpenCV, YOLOv8, Flask, React.js, TensorFlow, MongoDB
- Developed an AI-powered traffic management system across 4 roads using YOLOv8, OpenCV, Flask, and React.js for real-time vehicle detection and traffic analysis.
- Implemented an LSTM-based traffic prediction pipeline using 90 days of historical data with TensorFlow to forecast traffic patterns.
- Developed adaptive traffic signal optimization to dynamically adjust signal timings based on traffic density and improve traffic flow.

Accident Alert & Emergency Response System

- Technologies:** Java, Tesseract, GPS/Geo-tagging, MongoDB, Twilio.
- Developed an AI-powered accident management system that uses OCR-based number plate recognition and geo-tagging to detect accidents, identify vehicles, and automatically notify nearby hospitals and authorities for faster emergency response.

Certificates

- AI Literacy – IBM SkillsBuild
- Hands-on workshop on Git and GitHub
- Webworks workshop on HTML, CSS, JavaScript

